

Table 13: Detection performance on OpAI-Bench. Each entry reports AUROC / TPR@1% / TPR@0.1%, in percentage points. For train-based methods, results are mean $\pm$ standard deviation. “-” denotes training-free methods that do not use a source generator.

Train	Method	Gemini-2.5-Flash	GPT-5.4	GPT-5.4-nano	Qwen3-8B
-	Likelihood	71.42 / 0.10 / 0.00	49.45 / 0.03 / 0.00	71.37 / 0.15 / 0.00	59.45 / 0.04 / 0.00
	LRR	77.73 / 3.66 / 0.76	51.09 / 0.22 / 0.00	72.25 / 0.22 / 0.00	60.00 / 0.22 / 0.00
	Fast-DetectGPT	73.66 / 8.78 / 2.53	49.79 / 1.33 / 0.30	76.62 / 12.92 / 3.06	47.34 / 1.66 / 0.40
	Binoculars	52.12 / 0.07 / 0.00	59.18 / 1.66 / 0.43	36.85 / 0.07 / 0.00	74.54 / 3.98 / 0.63
	IRM	72.33 / 17.93 / 6.70	59.52 / 6.32 / 1.49	66.28 / 8.30 / 1.69	74.02 / 28.19 / 10.95
	Lastde++	67.88 / 4.66 / 1.01	46.62 / 1.46 / 0.54	71.03 / 6.98 / 1.85	41.79 / 0.46 / 0.00
	SpecDetect++	58.36 / 2.07 / 0.70	43.59 / 0.76 / 0.00	62.46 / 4.40 / 1.75	38.43 / 0.45 / 0.11
	NTS	57.81 / 2.03 / 0.73	51.36 / 1.36 / 0.30	51.91 / 2.00 / 0.20	51.36 / 1.68 / 0.40
Gemini	RoBERTa-L	100.00 $\pm$ 0.00 / 100.00 $\pm$ 0.00 / 99.44 $\pm$ 0.01	79.51 $\pm$ 0.03 / 36.11 $\pm$ 0.04 / 16.64 $\pm$ 0.07	87.10 $\pm$ 0.02 / 38.75 $\pm$ 0.04 / 15.88 $\pm$ 0.05	97.80 $\pm$ 0.01 / 76.00 $\pm$ 0.02 / 49.87 $\pm$ 0.04
	ImBD	81.06 $\pm$ 0.02 / 13.91 $\pm$ 0.02 / 10.70 $\pm$ 0.02	68.81 $\pm$ 0.01 / 3.87 $\pm$ 0.06 / 2.26 $\pm$ 0.05	47.36 $\pm$ 0.03 / 0.27 $\pm$ 0.02 / 0.10 $\pm$ 0.07	79.58 $\pm$ 0.02 / 12.59 $\pm$ 0.03 / 9.51 $\pm$ 0.05
	RepreGuard	97.82 $\pm$ 0.04 / 84.77 $\pm$ 0.02 / 77.07 $\pm$ 0.02	76.41 $\pm$ 0.03 / 17.12 $\pm$ 0.14 / 6.07 $\pm$ 0.12	84.98 $\pm$ 0.04 / 34.36 $\pm$ 0.04 / 21.01 $\pm$ 0.10	91.90 $\pm$ 0.03 / 57.30 $\pm$ 0.02 / 48.16 $\pm$ 0.02
	S2D	100.00$\pm$0.00 / 100.00$\pm$0.00 / 100.00$\pm$0.00	89.81$\pm$0.02 / 60.53$\pm$0.05 / 53.65$\pm$0.06	94.75$\pm$0.01 / 80.97$\pm$0.03 / 74.92$\pm$0.06	97.25 $\pm$ 0.00 / 92.92$\pm$0.01 / 89.40$\pm$0.03
GPT-5.4	RoBERTa-L	98.76$\pm$0.02 / 83.86$\pm$0.22 / 59.14$\pm$0.12	98.38 $\pm$ 0.01 / 70.28 $\pm$ 0.13 / 44.93$\pm$0.25	97.38 $\pm$ 0.02 / 69.62$\pm$0.21 / 41.85$\pm$0.26	95.32 $\pm$ 0.02 / 62.98 $\pm$ 0.19 / 41.22 $\pm$ 0.10
	ImBD	72.38 $\pm$ 0.03 / 2.34 $\pm$ 0.24 / 1.71 $\pm$ 0.10	87.99 $\pm$ 0.02 / 16.72 $\pm$ 0.15 / 11.03 $\pm$ 0.23	83.28 $\pm$ 0.03 / 9.07 $\pm$ 0.23 / 7.77 $\pm$ 0.28	83.43 $\pm$ 0.03 / 11.31 $\pm$ 0.21 / 6.39 $\pm$ 0.08
	RepreGuard	97.12 $\pm$ 0.02 / 80.75 $\pm$ 0.32 / 72.64$\pm$0.02	80.46 $\pm$ 0.16 / 21.09 $\pm$ 0.36 / 11.54 $\pm$ 0.20	87.89 $\pm$ 0.02 / 44.95 $\pm$ 0.01 / 29.18 $\pm$ 1.20	92.14 $\pm$ 0.52 / 57.42 $\pm$ 0.36 / 47.41 $\pm$ 1.02
	S2D	97.67 $\pm$ 0.01 / 92.13$\pm$0.23 / 71.58$\pm$0.11	98.56$\pm$0.00 / 81.04$\pm$0.14 / 38.35$\pm$0.24	98.36$\pm$0.01 / 68.29$\pm$0.22 / 49.94$\pm$0.27	96.47$\pm$0.01 / 73.09$\pm$0.20 / 56.38$\pm$0.09
GPT-5.4-nano	RoBERTa-L	99.29 $\pm$ 0.01 / 90.44 $\pm$ 0.01 / 83.32 $\pm$ 0.02	93.88 $\pm$ 0.05 / 61.28 $\pm$ 0.08 / 51.83 $\pm$ 0.03	99.67$\pm$0.01 / 95.04$\pm$0.02 / 90.77$\pm$0.02	97.17 $\pm$ 0.02 / 84.12 $\pm$ 0.02 / 78.35 $\pm$ 0.02
	ImBD	64.24 $\pm$ 0.02 / 1.99 $\pm$ 0.02 / 0.86 $\pm$ 0.03	83.40 $\pm$ 0.03 / 13.25 $\pm$ 0.10 / 7.78 $\pm$ 0.01	86.34 $\pm$ 0.02 / 19.91 $\pm$ 0.01 / 13.68 $\pm$ 0.03	86.53 $\pm$ 0.03 / 18.89 $\pm$ 0.03 / 14.05 $\pm$ 0.03
	RepreGuard	97.55 $\pm$ 0.02 / 82.15 $\pm$ 0.02 / 77.35 $\pm$ 0.02	81.13 $\pm$ 0.02 / 27.33 $\pm$ 0.12 / 23.64 $\pm$ 0.10	90.24 $\pm$ 0.04 / 56.27 $\pm$ 0.02 / 49.08 $\pm$ 0.02	92.69 $\pm$ 0.03 / 57.32 $\pm$ 0.03 / 52.09 $\pm$ 0.04
	S2D	99.97$\pm$0.00 / 99.47$\pm$0.00 / 98.57$\pm$0.01	95.93$\pm$0.04 / 70.32$\pm$0.09 / 59.93$\pm$0.02	99.64 $\pm$ 0.00 / 96.20$\pm$0.01 / 94.16$\pm$0.01	98.16$\pm$0.01 / 93.53$\pm$0.01 / 89.69$\pm$0.01
Qwen3-8B	RoBERTa-L	99.74 $\pm$ 0.01 / 97.04 $\pm$ 0.01 / 68.90 $\pm$ 0.02	89.82 $\pm$ 0.02 / 36.47 $\pm$ 0.04 / 10.37 $\pm$ 0.06	94.95 $\pm$ 0.02 / 64.28 $\pm$ 0.04 / 26.62 $\pm$ 0.08	99.42$\pm$0.01 / 94.38$\pm$0.01 / 75.63$\pm$0.02
	ImBD	72.73 $\pm$ 0.02 / 4.98 $\pm$ 0.02 / 4.67 $\pm$ 0.03	74.36 $\pm$ 0.01 / 4.00 $\pm$ 0.06 / 3.38 $\pm$ 0.08	64.85 $\pm$ 0.03 / 1.68 $\pm$ 0.02 / 1.36 $\pm$ 0.10	90.36 $\pm$ 0.02 / 32.81 $\pm$ 0.02 / 30.07 $\pm$ 0.03
	RepreGuard	96.53 $\pm$ 0.02 / 78.81 $\pm$ 0.02 / 62.18 $\pm$ 0.03	71.61 $\pm$ 0.12 / 11.71 $\pm$ 0.23 / 3.65 $\pm$ 0.12	80.13 $\pm$ 0.03 / 23.87 $\pm$ 0.01 / 8.29 $\pm$ 0.14	91.61 $\pm$ 0.02 / 61.85 $\pm$ 0.02 / 44.35 $\pm$ 0.02
	S2D	100.00$\pm$0.00 / 100.00$\pm$0.00 / 99.46$\pm$0.01	94.26$\pm$0.01 / 56.79$\pm$0.05 / 41.04$\pm$0.07	96.69$\pm$0.01 / 82.52$\pm$0.03 / 65.49$\pm$0.09	99.20 $\pm$ 0.00 / 91.36 $\pm$ 0.00 / 88.01$\pm$0.01

with the strongest probe. The advantage is particularly consistent under cross-generator evaluation: S2D achieves the highest OOD AUROC and TPR@1% for v4 and v8, while matching the best probe for v6. The gains are not uniform across all metrics; for example, the MLP achieves a higher ID TPR@1% on v4. Overall, directly fitting a discriminative classifier to the unsteered representations does not consistently recover the performance of S2D, particularly under cross-generator transfer. However, since S2D combines steering with a vMF-based readout whereas the probe baselines use unsteered representations with different classifiers, this comparison does not fully disentangle the individual contributions of steering and the downstream classifier.

G Omitted Theoretical Details and Proofs

G.1 Proof of Theorem 3.1

Proof of Theorem 3.1. Condition on the training sample $\mathcal{S}_{\text{train}}$, the estimated scoring function $\hat{\mathcal{S}}_t$ is fixed, and the only remaining randomness comes from the calibration sample $\mathcal{S}_{\text{cal}} = \{x_i^-\}_{i=1}^{n_2}$ with $x_i^- \stackrel{\text{i.i.d.}}{\sim} \mathbb{P}_0$.

Define the true survival function $\mathcal{T}(\tau) := \mathbb{P}_0(\hat{\mathcal{S}}_t(X_{\text{test}}^-) > \tau)$ (where X_{test}^- follows the same distribution of samples in $\mathcal{S}_{\text{cal}}$) and its empirical counterpart $\hat{\mathcal{T}}_{n_2}(\tau) := \frac{1}{n_2} \sum_{i=1}^{n_2} \mathbb{1}(\hat{\mathcal{S}}_t(x_i^-) > \tau)$. By the definition of the empirical threshold $\hat{\tau}_{\alpha,t}$ in Equation (5), it is the infimum of τ such that $\hat{\mathcal{T}}_{n_2}(\tau) \leq \alpha$. On one hand, by definition, we have $\hat{\mathcal{T}}_{n_2}(\hat{\tau}_{\alpha,t}) \leq \alpha$. On the other hand, by the continuity assumption, the calibration scores are almost surely distinct, so $\hat{\mathcal{T}}_{n_2}(\tau)$ is a step function whose jumps have size $1/n_2$. The infimum definition therefore guarantees that the empirical probability does not fall below α by more than $1/n_2$. Therefore, $\alpha - \hat{\mathcal{T}}_{n_2}(\hat{\tau}_{\alpha,t}) \leq \frac{1}{n_2}$. It implies that

$$\left| \hat{\mathcal{T}}_{n_2}(\hat{\tau}_{\alpha,t}) - \alpha \right| \leq \frac{1}{n_2}.$$

By applying the triangle inequality, we can bound the absolute difference between the population Type-I error and the target level α

$$\begin{aligned} |\mathcal{T}(\hat{\tau}_{\alpha,t}) - \alpha| &\leq \left| \mathcal{T}(\hat{\tau}_{\alpha,t}) - \hat{\mathcal{T}}_{n_2}(\hat{\tau}_{\alpha,t}) \right| + \left| \hat{\mathcal{T}}_{n_2}(\hat{\tau}_{\alpha,t}) - \alpha \right| \\ &\leq \sup_{\tau \in \mathbb{R}} \left| \mathcal{T}(\tau) - \hat{\mathcal{T}}_{n_2}(\tau) \right| + \frac{1}{n_2}. \end{aligned}$$

By using the Dvoretzky–Kiefer–Wolfowitz (DKW) inequality [80], the uniform deviation between the true and empirical functions is bounded by

$$\mathbb{P}\left(\sup_{\tau \in \mathbb{R}} |\mathcal{T}(\tau) - \hat{\mathcal{T}}_{n_2}(\tau)| > \sqrt{\frac{\log(2/\delta)}{2n_2}} \middle| \mathcal{S}_{\text{train}}\right) \leq 2e^{-2n_2\left(\sqrt{\frac{\log(2/\delta)}{2n_2}}\right)^2} = \delta.$$

Hence, conditional on $\mathcal{S}_{\text{train}}$, with probability at least $1 - \delta$, we have

$$|\mathcal{T}(\hat{\tau}_{\alpha,t}) - \alpha| \leq \sqrt{\frac{\log(2/\delta)}{2n_2}} + \frac{1}{n_2}.$$

Since $\mathbb{P}_0(\hat{\mathcal{S}}_t(X_{\text{test}}^-) > \hat{\tau}_{\alpha,t}) = \mathcal{T}(\hat{\tau}_{\alpha,t})$, we can rewrite the conditional probability as:

$$\mathbb{P}\left(\left|\mathbb{P}_0(\hat{\mathcal{S}}_t(X_{\text{test}}^-) > \hat{\tau}_{\alpha,t}) - \alpha\right| \leq \sqrt{\frac{\log(2/\delta)}{2n_2}} + \frac{1}{n_2} \middle| \mathcal{S}_{\text{train}}\right) \geq 1 - \delta,$$

which yields

$$\mathbb{P}\left(\left|\mathbb{P}_0(\hat{\mathcal{S}}_t(X_{\text{test}}^-) > \hat{\tau}_{\alpha,t}) - \alpha\right| \leq \sqrt{\frac{\log(2/\delta)}{2n_2}} + \frac{1}{n_2}\right) \geq 1 - \delta,$$

which completes the proof. $\square$

G.2 Formal Statement for Theorem 3.3

Theorem 3.3 characterizes the excess Type II error of the detector trained on $\mathcal{S}_{\text{train}}$. To establish this result, we introduce several additional technical assumptions. We first present and discuss these assumptions, followed by the formal statement of Theorem 3.3.

Two-Time-Scale Prototype Tracking. Algorithm 1 can be viewed as a two-time-scale stochastic approximation scheme for solving the empirical objective (6):

$$\mathcal{L}_{\text{vMF}}(\mathbf{v}, \boldsymbol{\mu}_0, \boldsymbol{\mu}_1) := \frac{1}{n_1} \sum_{i=1}^{n_1} \log q_{\beta}(y_i | f_{\theta, \mathbf{v}}(x_i), \boldsymbol{\mu}_0, \boldsymbol{\mu}_1), \quad (6)$$

where $\mathcal{S}_{\text{train}} = \{(x_i, y_i)\}_{i=1}^{n_1}$ consists of i.i.d. samples from $\mathcal{P}$.

We restate the key update steps for clarity and subsequent analysis. At iteration t , let $\mathbf{v}_t$ denote the current steering vector. For each class c , define

$$\bar{\mathbf{z}}_{c,t}(\mathbf{v}_t) = \frac{1}{|\{(x, y) : x \in \mathcal{B}_t, y = c\}|} \sum_{(x,y): x \in \mathcal{B}_t, y=c} f_{\theta, \mathbf{v}_t}(x)$$

as the average representation of class c over a fresh mini-batch $\mathcal{B}_t$. The update rule is given by

$$\textbf{Fast Process (Mean Direction Tracking)} \quad \hat{\boldsymbol{\mu}}_{c,t+1} = \Pi_{\mathbb{S}^{d-1}}((1 - \rho)\hat{\boldsymbol{\mu}}_{c,t} + \rho \bar{\mathbf{z}}_{c,t}(\mathbf{v}_t)), \quad (7a)$$

$$\textbf{Slow Process (Steering Vector Update)} \quad \mathbf{v}_{t+1} = \mathbf{v}_t + \eta \nabla_{\mathbf{v}} \mathcal{L}_{\text{vMF}}(\mathbf{v}_t, \hat{\boldsymbol{\mu}}_{0,t}, \hat{\boldsymbol{\mu}}_{1,t}), \quad (7b)$$

where $\Pi_{\mathbb{S}^{d-1}}(\mathbf{x}) = \mathbf{x}/\|\mathbf{x}\|$ denotes projection onto the unit sphere. Here, $\hat{\boldsymbol{\mu}}_{c,t}$ serves as an estimator of the population mean direction for class c at iteration t . We consider a two-time-scale regime with $0 < \eta \ll \rho \leq 1$, under which $\{\hat{\boldsymbol{\mu}}_{c,t}\}_{c=0}^1$ evolves on the fast timescale while $\mathbf{v}_t$ evolves on the slow timescale.

Assumptions. Assumption 1 introduces a generative model under which the representation follows a vMF distribution at the population optimum. This assumption facilitates the theoretical analysis.

Assumption 1 (Local vMF Model). For every $\mathbf{v}$ and each class $c \in \{0, 1\}$, define $Z_{\mathbf{v}} = f_{\theta, \mathbf{v}}(X)$ and assume

$$Z_{\mathbf{v}} | Y = c \sim \text{vMF}(\boldsymbol{\mu}_c(\mathbf{v}), \kappa),$$

that is, its density is given by $p(z | Y = c) = C_d(\kappa) \exp(\kappa \boldsymbol{\mu}_c(\mathbf{v})^\top z)$ for any $z \in \mathbb{S}^{d-1}$, where $C_d(\kappa)$ is the normalization constant.

Remark G.1 (Gradient Noises). Let $\mathcal{F}_t$ denote the filtration generated by the optimization trajectory up to iteration t . Under Assumption 1, the conditional expectation of the mini-batch average satisfies

$$\mathbb{E}[\bar{\mathbf{z}}_{c,t}(\mathbf{v}_t) \mid \mathcal{F}_t] = A_d(\kappa) \boldsymbol{\mu}_c(\mathbf{v}_t),$$

where $A_d(\kappa) = \frac{I_{d/2}(\kappa)}{I_{d/2-1}(\kappa)}$, and $I_\nu(\cdot)$ denotes the modified Bessel function of the first kind of order ν . Based on the observation, we can define the following gradient noise

$$\boldsymbol{\zeta}_{c,t+1} := \bar{\mathbf{z}}_{c,t}(\mathbf{v}_t) - A_d(\kappa) \boldsymbol{\mu}_c(\mathbf{v}_t), \quad (8)$$

for a class $c \in \{0, 1\}$. We then have that $\mathbb{E}[\boldsymbol{\zeta}_{c,t+1} \mid \mathcal{F}_t] = 0$ and $\|\boldsymbol{\zeta}_{c,t+1}\| \leq 2$ almost surely due to the fact that $\|\bar{\mathbf{z}}_{c,t}(\mathbf{v}_t)\|$, $A_d(\kappa)$, and $\|\boldsymbol{\mu}_c(\mathbf{v}_t)\| \leq 1$.

Assumption 2 (Local Tracking Conditions). Define $\boldsymbol{\mu}_{c,t} := \boldsymbol{\mu}_c(\mathbf{v}_t)$ and assume that

1. **(Bounded drift)** There exists $C_\mu > 0$ such that $\|\boldsymbol{\mu}_{c,t+1} - \boldsymbol{\mu}_{c,t}\| \leq C_\mu \eta$.
2. **(Initialization)** $\|\hat{\boldsymbol{\mu}}_{c,0} - \boldsymbol{\mu}_{c,0}\| \leq 1/4$.

Theorem G.2 (Local Mean Direction Tracking). Fix $c \in \{0, 1\}$ and write $\boldsymbol{\mu}_{c,t} := \boldsymbol{\mu}_c(\mathbf{v}_t)$. Consider the recursion

$$\mathbf{z}_{c,t+1} = (1 - \rho) \hat{\boldsymbol{\mu}}_{c,t} + \rho \bar{\mathbf{z}}_{c,t}(\mathbf{v}_t), \quad \hat{\boldsymbol{\mu}}_{c,t+1} = \frac{\mathbf{z}_{c,t+1}}{\|\mathbf{z}_{c,t+1}\|}.$$

Under Assumptions 1 and 2, there exist constants $c, C, \rho_0, \gamma_0 > 0$ such that, if

$$0 < \rho \leq \rho_0, \quad \frac{\eta}{\rho} \leq \gamma_0, \quad \rho \log \frac{2T}{\delta} \leq 1,$$

and

$$\|\hat{\boldsymbol{\mu}}_{c,0} - \boldsymbol{\mu}_{c,0}\|^2 + C \left(\sqrt{\rho \log \frac{2T}{\delta}} + \frac{\eta^2}{\rho^2} \right) \leq \frac{1}{16},$$

then, with probability at least $1 - \delta$,

1. For all $0 \leq t \leq T$, $\|\hat{\boldsymbol{\mu}}_{c,t} - \boldsymbol{\mu}_{c,t}\|^2 \leq (1 - c\rho)^t \|\hat{\boldsymbol{\mu}}_{c,0} - \boldsymbol{\mu}_{c,0}\|^2 + C \left(\sqrt{\rho \log \frac{2T}{\delta}} + \frac{\eta^2}{\rho^2} \right)$.
2. For all $0 \leq t \leq T$, $\|\hat{\boldsymbol{\mu}}_{c,t} - \boldsymbol{\mu}_{c,t}\| \leq \frac{1}{4}$, $\|\mathbf{z}_{c,t+1}\| \geq \frac{1}{4}$, $\langle \hat{\boldsymbol{\mu}}_{c,t}, \boldsymbol{\mu}_{c,t} \rangle > 0$.

Theorem G.2 establishes that the proposed exponential moving average update $\hat{\boldsymbol{\mu}}_{c,t}$ can accurately track the evolving class prototypes $\boldsymbol{\mu}_{c,t}$ under the local vMF model. Despite stochastic gradient noise and slow drift in the true mean direction, the estimated prototype remains close to the target with high probability. The result shows a contraction behavior up to a steady-state error, where the averaging parameter and the drift rate govern the tracking accuracy. Its proof is deferred in Appendix G.4.

Since the score function depends on the class prototypes only through their difference, the tracking error bound in Theorem G.2 directly translates into a uniform bound on the deviation between the plug-in and oracle score functions (i.e., $\hat{\mathcal{S}}_t$ and $\mathcal{S}_t$), as formalized in the following corollary.

Corollary G.3. Define the oracle and plug-in log-likelihood ratios with respect to the same steered representation $f_{\theta, \mathbf{v}_t}(x)$ as

$$\mathcal{S}_t(x) = \kappa(\boldsymbol{\mu}_{1,t} - \boldsymbol{\mu}_{0,t})^\top f_{\theta, \mathbf{v}_t}(x), \quad \hat{\mathcal{S}}_t(x) = \kappa(\hat{\boldsymbol{\mu}}_{1,t} - \hat{\boldsymbol{\mu}}_{0,t})^\top f_{\theta, \mathbf{v}_t}(x).$$

Under Assumptions 1 and 2, for any $\delta \in (0, 1)$, suppose that the conditions of Theorem G.2 hold for both classes with failure probability $\delta/2$. Then, with probability at least $1 - \delta$, there exist constants $c \in (0, 1)$ and $C > 0$ such that the following uniform bound holds for all $0 \leq t \leq T$:

$$\sup_{x \in \mathcal{X}} |\hat{\mathcal{S}}_t(x) - \mathcal{S}_t(x)| \leq r_t(\delta, \rho, \eta),$$

where

$$r_t(\delta, \rho, \eta) := \kappa C \sqrt{(1 - c\rho)^t + \sqrt{\rho \log \frac{4T}{\delta}} + \frac{\eta^2}{\rho^2}}.$$